École Nationale des Ponts et Chaussées

Département de première année

Bases scientifiques pour la transition énergétique 1

Séance 3 : Récapitulatif des points essentiels partie rayonnement

2026

Recap of what must be known for the exam

1. Planck's law

The spectral density of a blackbody at temperature T is :

$$\mathcal{B}_{\hbar\omega,\Omega}(T) = \frac{\cos(\theta)}{\pi} B_{\hbar\omega}(T) \text{ with } B_{\hbar\omega}(T) = \frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2}\left(\exp\left(\frac{\hbar\omega}{k_BT}\right)-1\right)^{-1}$$

$\mathcal{B}_{\hbar\omega,\Omega}(T)\, d(\hbar\omega)\, d\Omega d(\hbar\omega)$ represents the number of photons with energy between $\hbar\omega$ and $\hbar\omega + d(\hbar\omega)$ emitted per unit time and per unit surface in the directions between Ω and $\Omega + d\Omega$. $B_{\hbar\omega}(T)$ is the spectral density emitted in all direction.

The power density $\mathcal{P}_{\hbar\omega,\Omega}(T) = \hbar\omega\mathcal{B}_{\hbar\omega,\Omega}(T)$ represents the power emitted (since $\hbar\omega$ is the energy of a photon). Integration of $\mathcal{P}_{\hbar\omega,\Omega}$ over all energies and all directions give the total power emitted per unit surface of a blackbody (Stefan-Boltzmann law) :

$$\mathcal{P}(T) = \int_0^{+\infty} \hbar\omega\mathcal{B}_{\hbar\omega}(T)\, d(\hbar\omega) = \sigma T^4 \text{ with } \sigma = \frac{\pi^4}{15}\frac{k_B^4}{4\pi^2\hbar^3c^2} \text{ the Stefan-Boltzmann constant}$$

Photons can be described equivalently in terms of their energy $\hbar\omega$, frequency $\nu = \frac{\omega}{2\pi}$, or wavelength $\lambda = \frac{c}{\nu}$. Therefore, Planck's law can also be expressed in terms of ν or λ. The respective spectral densities have different units and satisfy:

$$|\mathcal{B}_{\hbar\omega,\Omega} d(\hbar\omega)| = |\mathcal{B}_{\nu,\Omega}(T)\, d\nu| = |\mathcal{B}_{\lambda,\Omega}(T)\, d\lambda|$$

2. Kirchhoff's law

The absorptivity $\alpha(\hbar\omega,\Omega)$ is the fraction of incident photons absorbed at frequency ω from an incident direction Ω). The emissivity $\varepsilon(\hbar\omega,\Omega)$ is the ability of the body to emit a radiation at frequency ω in the direction Ω relative to the emission of a blackbody.

$$\text{Kirchhoff's law: } \alpha(\hbar\omega,\Omega) = \varepsilon(\hbar\omega,\Omega)$$

For a blackbody, absorptivity and emissivity are equal to 1 at all frequencies. A material with absorptivity and emissivity < 1 is called a grey body.

3. Solar resource

Applying the Stefan-Boltzmann law, the total radiation power emitted by the Sun is $\mathcal{P}_\odot = 4\pi R_\odot^2 \times \sigma T_\odot^4$, with $R_\odot$ the radius of the Sun, and $T_\odot = 5\,800$K the surface temperature of the Sun. The power per unit surface received on Earth at a distance $D_{T-\odot}$ in ideal conditions (surface perfectly oriented, before the atmosphere) is therefore $\phi_{\text{AM }0} = \frac{\mathcal{P}_\odot}{4\pi(D_{T-\odot})^2} = 1\,360$ W/m^2 (called solar constant). The total power received from the Sun on Earth is 10 000 times larger than humanity's energy needs.

Taking into account the atmosphere, the seasonal variations and the day/night cycles, the actual power received on average at the surface of the Earth is about 10 times less than the above estimate, e.g., 150 W/m^2 in France.

A more accurate calculation of the solar power received per unit surface is:

$$\phi = (\sin(\theta))^2 \sigma T_\odot^4$$

where θ is the angle occupied by the radius of the Sun in the sky. In absence of concentration $\sin(\theta) = \sin(\theta_S) = R_\odot / D_{T-\odot}$ and one recovers the AM 0 estimate (1 360 W/m^2). With a concentration system, θ can be increased : $(\sin(\theta))^2 = C(\sin(\theta_S))^2$, with $C \geq 1$ the concentration factor. For the maximum concentration ($\sin(\theta) = 1$) the solar power received can be increased by a factor $C = 46\,000$.

3. Solar thermal energy

A thermal solar panel absorbs radiation from the Sun, remits a radiation that depends on its temperature because of its emissivity, and provides heat for subsequent use (e.g., domestic heating). In stationary conditions, the efficiency is obtained from the energy balance and is displayed in Figure **??**. This efficiency falls to zero as the temperature of the panel approaches the maximum temperature $T_p = \sqrt{\sin(\theta_S)} T_\odot = 122$°C. Turning the heat into work proves very inefficient (efficiency $< 6\%$), even with an ideal Carnot engine, because the temperature of the panel remains small. Solar concentration is necessary to reach a reasonable efficiency.

4. Solar photovoltaic

Photovoltaic pannels are made of semiconductors that are greybody, which absorb/emit radiation only for frequencies above a given threshold $\hbar\omega > E_{gap}$ (energy gap). Accordingly, $\alpha(\hbar\omega) = \varepsilon(\hbar\omega) = 1$ if $\hbar\omega > E_{gap}$ and $\alpha(\hbar\omega) = \varepsilon(\hbar\omega) = 0$ otherwise. The semiconductor is chosen so that the energy gap E_{gap} allows most of most of the solar spectrum to be absorbed, while most of blackbody emission at the temperature of the panel is avoided.

One can estimate an upper bound of the efficiency of a photovoltaic pannel by assuming that all electrons excited by a photon of energy $> E_{gap}$ contribute to the current collected (no loss of current), and that the electrical energy provided by each electron is exactly E_{gap} (no loss of voltage). One obtains the so-called Trivich-Finn limit, which reaches a maximum of 43% for a energy gap of 1.1 eV (Fig. **??**) :

$$\eta = E_{gap}(\sin(\theta_S))^2 \int_{E_{gap}}^{+\infty} \frac{(\hbar\omega)^2}{4\pi^2\hbar^3c^2} \left(\exp\left(\frac{\hbar\omega}{k_B T_\odot}\right) - 1\right)^{-1} d(\hbar\omega)$$

A more accurate estimation of the efficiency is given by the Shockley-Queisser limit, which takes into account the loss of current to the electrons recombining before being collected, and the loss of voltage required to collect the electrons. The Shockley-Queisser limit reaches a maximum of about 30% for a gap of 1.25 eV (Fig. **??**). Most solar cells are made of silicon, a widely available resource, with an energy gap of 1.1 eV close to the optimal gap .